

ANSWER KEY

SAT Practice Test 1 - Standardized Edition

Section 1: Reading and Writing

Module 1 | Module 2 Easier | Module 2 Harder

Section 2: Math

Module 1 | Module 2 Easier | Module 2 Harder

Answers and explanations are preserved from the original source file.

Section 1: Reading and Writing

Module 1 (27 Questions) - All Students

Time: 32 minutes

Q1 | Correct Answer: B Skill: Central Idea | Difficulty: Medium

Why B is correct: The passage states that Utamaro “earned wide recognition” for his portraiture work in *bijin-ga*, that his “commitment to this subject was so influential” that later artists worked “in the manner of Utamaro.” This directly supports the idea that he played a significant role in shaping the portraiture tradition.

Trap analysis: - A) Wrong detail - the passage says Utamaro focused on portraits, not landscapes. This reverses a detail. - C) Too extreme - the passage says Utamaro mastered *bijin-ga*, not that he introduced it. The tradition existed before him. - D) Unsupported - the passage says later artists imitated Utamaro, not that they surpassed him.

Q2 | Correct Answer: A Skill: Boundaries/Punctuation (Colon) | Difficulty: Medium

Why A is correct: The sentence before the blank makes a complete statement (“Both the accessibility... cycle matter”). What follows is an explanation of how they matter. A colon is the correct punctuation to introduce an explanatory clause after an independent clause.

Trap analysis: - B) No punctuation creates a run-on sentence. - C) A comma creates a comma splice between two independent clauses. - D) A comma after “matter” doesn’t properly separate the clauses.

Q3 | Correct Answer: A Skill: Command of Evidence (Textual) | Difficulty: Medium

Why A is correct: The claim is that Lily is “acutely aware of the financial pressures that shape her choices.” Choice A describes her as “the victim of the

civilization which had produced her” with bracelet links seeming “like manacles chaining her to her fate”-directly illustrating awareness of societal/financial constraints.

Trap analysis: - B) Describes Mrs. Peniston’s house, not financial pressure. - C) Shows Lily reminiscing and looking out a window - no connection to financial awareness. - D) Describes domestic details of a household, not financial pressures.

Q4 | Correct Answer: A Skill: Transitions | Difficulty: Easy

Why A is correct: The sentence before the blank describes how organisms have “specialized proteins.” The sentence after gives a specific example of how these proteins work. “For instance” correctly introduces a specific example of a general claim.

Trap analysis: - B) “In contrast” signals opposition, but the second sentence supports the first. - C) “Regardless” dismisses what came before, which is illogical here. - D) “As a result” implies causation, but the second sentence illustrates rather than results from the first.

Q5 | Correct Answer: B Skill: Rhetorical Synthesis | Difficulty: Medium

Why B is correct: The goal is to specify when the axolotl population was rediscovered. Choice B includes both the year (2013) and the location (Xochimilco), directly fulfilling the stated goal.

Trap analysis: - A) Mentions habitat loss but not the rediscovery date. - C) Discusses pangolins, not axolotls - wrong species for the goal. - D) Incorrectly states both were “once thought to be extinct” (this is not stated about pangolins).

Q6 | Correct Answer: C Skill: Text Purpose | Difficulty: Medium

Why C is correct: The passage explains that Morrison’s narratives “resist reducing characters to symbols of suffering” and grant them “desires, contradictions, and moments of joy.” This describes a balance between historical themes and full characterization.

Trap analysis: - A) The passage is about characterization technique, not tracing historical causes. - B) Directly contradicts the passage - Morrison resists reducing characters to symbols. - D) The passage doesn’t compare Morrison to other novelists.

Q7 | Correct Answer: C Skill: Text Purpose | Difficulty: Easy

Why C is correct: The passage describes how Siqueiros used industrial paints and spray guns, which “expanded the possibilities available to public artists.” The main purpose is describing how his unconventional materials broadened public mural art.

Trap analysis: - A) Only Siqueiros is discussed - no contrast with other muralists. - B) Too extreme - the text doesn’t claim industrial paints are superior “in all artistic applications.” - D) The text doesn’t claim spray guns “replaced” brushwork everywhere.

Q8 | Correct Answer: A Skill: Text Structure | Difficulty: Hard

Why A is correct: The text first identifies the ecological value of reefs (less than 1% of ocean floor, 25% of marine species), then describes a threat (coral bleaching from warming), then outlines the needed response (both local conservation and global emissions action).

Trap analysis: - B) There is no debate between competing positions - only Parmesan’s view is presented. - C) No conservation policy is critiqued; the text describes a problem and needed response. - D) The text focuses on one ecosystem (reefs) and one primary threat (warming), not “several unrelated threats.”

Q9 | Correct Answer: B Skill: Vocabulary in Context | Difficulty: Easy

Why B is correct: The passage says the novels do not “simply record” events but instead capture “emotional and psychological shifts.” “Trace” means to follow or track something through its development, which fits the idea of following emotional changes.

Trap analysis: - A) “Obscure” means to hide - opposite of what the novels do. - C) “Challenge” implies opposition, but the novels capture shifts, not challenge them. - D) “Predict” implies forecasting; the novels depict what happens, not predict it.

Q10 | Correct Answer: C Skill: Vocabulary in Context | Difficulty: Easy

Why C is correct: The passage describes selective breeding that changed the fruit from small, tough, and tart to large, thin-skinned, and mild. “Altering” (changing) the original characteristics is the precise description of this process.

Trap analysis: - A) “Reinforcing” means strengthening - the opposite of what happened (characteristics were changed, not reinforced). - B) “Cataloging”

means recording or listing - selective breeding changes characteristics, not catalogs them. - D) “Concealing” means hiding - the characteristics were changed, not hidden.

Q11 | Correct Answer: A Skill: Inference (Literary) | Difficulty: Medium

Why A is correct: Janie has “glossy leaves and bursting buds” (metaphor for youth/potential) and “wanted to struggle with life but it seemed to elude her.” She searches from “the top of the front steps” and then the “front gate,” gazing “up and down the road” - all indicating restless longing for something beyond her current setting.

Trap analysis: - B) “Nothing on the place nor in her grandmama’s house answered her” - she is not satisfied. - C) Nothing in the text indicates fear; she is curious and longing. - D) No specific past decision is referenced.

Q12 | Correct Answer: C Skill: Vocabulary in Context | Difficulty: Medium

Why C is correct: The passage says the zigzag route lets birds “take advantage of prevailing wind patterns,” and despite “increasing its total distance,” there is an effect on energy expenditure. The logical word is “reducing” - the wind assistance decreases energy costs.

Trap analysis: - A) “Maximizing” energy expenditure is the opposite of the benefit described. - B) “Documenting” doesn’t fit - the sentence describes an effect of the route, not a recording action. - D) “Overlooking” doesn’t logically complete the sentence.

Q13 | Correct Answer: A Skill: Command of Evidence (Strengthening) | Difficulty: Hard

Why A is correct: The underlined claim states that “some theropods with seemingly carnivorous teeth showed wear patterns consistent with a plant-heavy diet.” Finding that serrated-teeth theropods had scratch patterns matching herbivorous iguanas directly supports this claim.

Trap analysis: - B) Information about habitats of peg-shaped-teeth theropods doesn’t address the underlined claim about carnivorous-looking teeth with herbivore wear. - C) Uniform crocodilian patterns don’t tell us about theropod dietary diversity. - D) The number of tooth shapes over time doesn’t speak to the specific claim about mismatched form and diet.

Q14 | Correct Answer: B Skill: Cross-Text Connection | Difficulty: Medium

Why B is correct: Text 2's Chen argues that reading women's travel writing "solely through the lens of personal empowerment overlooks" how these texts "reinforced colonial ideologies." He would respond to Singh by pointing out that her focus on liberation ignores the colonial perspectives.

Trap analysis: - A) Chen doesn't discuss readership size. - C) Chen argues these texts reinforced colonial ideologies - he would not say they "challenged" them. - D) Chen doesn't compare male and female travel writers in terms of who was more critical of colonialism.

Q15 | Correct Answer: C Skill: Inference/Logical Completion | Difficulty: Medium

Why C is correct: The passage establishes that repeated bark stripping by "growing elephant populations" kills baobabs. The logical completion is that protection may depend on managing elephant density in areas with ancient trees.

Trap analysis: - A) Understanding why elephants prefer baobab bark is interesting but doesn't address the protection need. - B) Directly contradicts the passage - repeated stripping kills the oldest trees. - D) "Exclusively" relocating elephants is too extreme and impractical.

Q16 | Correct Answer: A Skill: Boundaries (Relative Clause) | Difficulty: Easy

Why A is correct: "Where" introduces a nonrestrictive relative clause describing what happens in the forested areas. A comma before "where" is standard for nonrestrictive clauses.

Trap analysis: - B) Comma after "where" is misplaced. - C) A period before "Where" creates a fragment. - D) No comma before a nonrestrictive "where" clause is incorrect.

Q17 | Correct Answer: D Skill: Boundaries (Appositive) | Difficulty: Easy

Why D is correct: "Samuel Beckett" is a compound subject with "Alberto Giacometti." No punctuation should separate the second subject from the shared verb "explored."

Trap analysis: - A) A comma after "Beckett" incorrectly separates the subject from the verb. - C) A semicolon is wrong - what follows is not an independent clause. - B) A colon doesn't fit because no list or explanation follows directly.

Q18 | Correct Answer: B Skill: Boundaries (Comma with Appositive) | Difficulty: Medium

Why B is correct: “Penicillin” is followed by an appositive phrase (“what would become one of the most important medical advances”). A comma properly introduces this appositive.

Trap analysis: - A) A colon would work if “penicillin” ended an independent clause, but “development of penicillin” needs the appositive to complete the thought. - C) No punctuation creates a confusing run-on. - D) A dash could work stylistically but the comma is the standard convention here.

Q19 | Correct Answer: A Skill: Boundaries (Nonrestrictive Element) | Difficulty: Medium

Why A is correct: “A facility built into the permafrost of a Norwegian mountainside” is a nonrestrictive appositive describing the Svalbard Global Seed Vault. It must be set off by commas - one after “Vault” (implied by the sentence structure) and one after “mountainside.”

Trap analysis: - C) A semicolon is wrong - what follows is the continuation of the same clause. - B) A colon doesn’t fit the grammatical structure. - D) No punctuation fails to close the nonrestrictive appositive.

Q20 | Correct Answer: C Skill: Vocabulary in Context | Difficulty: Medium

Why C is correct: The passage says that readers “who aren’t familiar with Spanish can easily follow the narrative” because the English text around the Spanish words reveals their meaning. “Illuminate” means to clarify or shed light on - matching the context perfectly.

Trap analysis: - A) “Contradict” would mean the English opposes the Spanish meaning - nonsensical. - B) “Obscure” means to hide - the opposite of the intended meaning. - D) “Exaggerate” means to overstate - not logically connected to providing meaning.

Q21 | Correct Answer: D Skill: Transitions | Difficulty: Medium

Why D is correct: The first sentence discusses R.U.R.’s “linguistic contribution” (coining “robot”). The second sentence describes an additional achievement (deepening a speculative fiction tradition). “Beyond this linguistic contribution” logically introduces an additional point.

Trap analysis: - A) “As a consequence” implies causation - the tradition isn’t a consequence of coining a word. - C) “In spite of” implies the second point works against the first, which is illogical. - B) “Regardless of the play’s reception” introduces irrelevant information about reception.

Q22 | Correct Answer: A Skill: Inference/Logical Completion | Difficulty: Hard

Why A is correct: The passage notes that captive dolphins (no predators) still exhibit unihemispheric sleep. Johansson uses this to argue against predator avoidance as the primary driver, suggesting the need to breathe at the surface is the primary reason.

Trap analysis: - B) The passage doesn’t compare total sleep hours. - C) The passage doesn’t claim unihemispheric sleep is unique to dolphins. - D) This contradicts the logic - breathing is the function supported, not diminished.

Q23 | Correct Answer: D Skill: Inference (Literary) | Difficulty: Medium

Why D is correct: The narrator notes “the same high cheekbones, the same watchful eyes” (physical features unchanged) “and yet something fundamental had shifted” - a “new posture, a new cadence of speech, a new way of occupying space.” The woman looks the same but presents herself differently.

Trap analysis: - A) Contradicts the text - the face is “unmistakably the same.” - B) The text doesn’t suggest deliberate concealment of identity. - C) Irene recognizes the woman - the “familiarity” is clear.

Q24 | Correct Answer: C Skill: Boundaries (Dash/Appositive) | Difficulty: Hard

Why C is correct: The clause “a finding that surprised researchers...” is a nonrestrictive appositive explaining the preceding statement. A dash effectively introduces this explanatory aside, especially after “ancestors.”

Trap analysis: - A) A comma works but doesn’t create the emphatic pause that the long explanatory aside warrants. - B) A colon would need the preceding text to be a complete independent clause. - D) No punctuation creates a confusing run-on.

Q25 | Correct Answer: B Skill: Rhetorical Synthesis | Difficulty: Medium

Why B is correct: The goal is to emphasize the railway’s record-setting length AND provide historical context. Choice B includes the length (9,289 km), the route (Moscow to Vladivostok), historical context (completed in 1916, 25 years), and the record (longest in the world).

Trap analysis: - A) Discusses the Qinghai-Tibet Railway, not the Trans-Siberian. - C) Too vague - doesn’t specify the length or historical details. - D) Compares the two railways but doesn’t provide the completion date or construction duration.

Q26 | Correct Answer: C Skill: Rhetorical Synthesis | Difficulty: Hard

Why C is correct: The goal is to acknowledge a complication in attribution. Choice C mentions Lovelace’s 1843 notes, the Bernoulli algorithm, AND the ongoing debate about whether Lovelace or Babbage deserves credit.

Trap analysis: - A) Simply states facts without addressing the attribution complication. - B) Focuses on the Analytical Engine, not the attribution debate. - D) Describes what the program computed but doesn’t mention the complication.

Q27 | Correct Answer: C Skill: Rhetorical Synthesis | Difficulty: Medium

Why C is correct: The goal requires describing Maathai’s approach AND explaining its broader significance. Choice C describes her method (organizing women to plant trees while advocating for rights) and cites Osei’s analysis of its broader significance (unified environmental-social cause).

Trap analysis: - A) States facts but doesn’t explain significance. - B) Paraphrases Osei without connecting it to Maathai’s specific work. - D) Mentions the Nobel Prize but doesn’t describe her approach or its significance.

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Section 1: Reading and Writing

Module 2 - Easier Path (27 Questions)

Time: 32 minutes

Q1 | **Correct Answer: A** Monarch butterflies migrate long distances using navigational methods (sun + magnetic field). B is too extreme (“only insect”). C says scientists disagree (not stated). D is unsupported.

- Q2 | Correct Answer: B** Goal: identify Soyinka + significance. B gives his nationality, roles, and the “first African” milestone.
- Q3 | Correct Answer: D** O’Keeffe turns tiny details into monumental compositions - “transform” fits precisely.
- Q4 | Correct Answer: C** The undocumented patterns led Vega to a conclusion - “On the basis of this evidence” is the logical transition.
- Q5 | Correct Answer: A** Jo says “That’s why I do it” after being told whistling is “boyish” - she deliberately defies convention.
- Q6 | Correct Answer: C** The sentence contrasts Gutenberg’s invention with earlier technologies. “However” signals this contrast.
- Q7 | Correct Answer: D** “As Arctic temperatures rise and sea ice recedes, polar bears must...” - a comma after “recedes” correctly continues the introductory adverbial clause.
- Q8 | Correct Answer: A** Two independent clauses joined by “and” with a comma: “connected by approximately 400 bridges, and the city’s unusual geography...”
- Q9 | Correct Answer: A** “Which blended elements of blues, gospel, and classical music,” - the comma closes the nonrestrictive relative clause.
- Q10 | Correct Answer: C** “Throughout the narrative, Odysseus encounters...” - a comma after “narrative” completes the introductory phrase.
- Q11 | Correct Answer: B** “...limitations ecologist Marco Torres and his team discovered while reviewing...” - no punctuation needed; “while” subordinates the clause.
- Q12 | Correct Answer: A** “...one of the driest places on Earth that scientists have studied...” - “places” flows directly into the restrictive clause with no punctuation.
- Q13 | Correct Answer: C** Rotating between nitrogen-fixing crops and grains decreases the need for fertilizers - “reduce” is correct.
- Q14 | Correct Answer: B** The detection systems detect changes, and as a consequence they provide evacuation time - “As a result.”
- Q15 | Correct Answer: A** The narrator walks slowly “to enjoy and analyse the species of pleasure” - she finds satisfaction in solitary observation.
- Q16 | Correct Answer: B** “That knowledge settled over her like a warm cloth” - the simile indicates gentle envelopment, not weighing down.
- Q17 | Correct Answer: A** The text describes how bees share navigational info through the waggle dance.
- Q18 | Correct Answer: B** Stones fit so tightly that a knife blade cannot be “inserted” between them.

- Q19 | Correct Answer: D** The anemone eats the prey attracted by the clownfish - “consumes.”
- Q20 | Correct Answer: D** Two independent clauses: “...angular geometries; her distinctive style challenged...” - a semicolon joins them.
- Q21 | Correct Answer: B** Park values traditional methods but says they need modern tech to meet current demands.
- Q22 | Correct Answer: A** The text describes Al-Khazneh and clarifies it was likely a tomb, not a treasury.
- Q23 | Correct Answer: B** Blood redirected to core → extremities receive less blood. “Consequently” shows this result.
- Q24 | Correct Answer: D** Edison “opposed” Tesla’s AC system, promoting DC instead.
- Q25 | Correct Answer: C** “Found in the rainforests of Southeast Asia, the plant...” - comma after “Asia” closes the introductory participial phrase.
- Q26 | Correct Answer: B** The claim is that Mole is “captivated by the unfamiliar world above ground.” Choice B describes his happiness upon seeing the river for the first time.
- Q27 | Correct Answer: B** Goal: ecological importance + threat. B mentions 1,500 fish species (importance) and 50% bleaching damage (threat).

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Section 1: Reading and Writing

Module 2 - Harder Path (27 Questions)

Time: 32 minutes

Q1 | **Correct Answer: C** The models show tonal contrasts are more perceptible at distance - this supports Takahashi's hypothesis.

Q2 | **Correct Answer: B** Isabel enjoys "great crowded streets" and "pushing her way through knots of loungers" because it gives her "a taste of the world's immensity" - she's drawn to bustling environments.

- Q3** | **Correct Answer: A** The limitation is that MRI conditions differ from “natural language use” - lying still in a scanner isn’t like spontaneous conversation.
- Q4** | **Correct Answer: B** Reyes argues Okafor’s reading is “incomplete” because it overlooks *vanitas* symbols that complicate a purely celebratory interpretation.
- Q5** | **Correct Answer: B** Critics contend the components wouldn’t provide survival advantages individually, making gradual evolution implausible - the classic irreducible complexity objection.
- Q6** | **Correct Answer: D** Resource wealth enables patronage networks, which weaken democratic accountability - “undermining.”
- Q7** | **Correct Answer: A** “...used to seal ceramic containers, raising the possibility...” - a comma introduces the participial phrase.
- Q8** | **Correct Answer: A** No difference in switching speed between groups → grammatical similarity doesn’t meaningfully affect switching speed.
- Q9** | **Correct Answer: D** “Consistent with those of known natural languages-” - a dash introduces the explanatory aside about why this matters.
- Q10** | **Correct Answer: C** The dashes may reflect “editorial decisions rather than [the author’s] intentions” - “authorial” is correct.
- Q11** | **Correct Answer: D** Satellite data can be misleading → as a result, images can mask biodiversity decline.
- Q12** | **Correct Answer: A** Digital platforms transform political “participation” into algorithm-driven content creation.
- Q13** | **Correct Answer: A** “Translated into Latin two centuries earlier, making it available...” - comma after “earlier” before the participial phrase.
- Q14** | **Correct Answer: C** Assigning monetary values risks reducing/lessening intrinsic value - “diminishing.”
- Q15** | **Correct Answer: B** Goal: define nonlinear narration + explain why Morrison used it. B does both.
- Q16** | **Correct Answer: D** “...conditions are unavoidable, such as textile dyeing...” - comma introduces exemplification.
- Q17** | **Correct Answer: A** The heat, dust, and stench “worked painfully upon” his “already overwrought nerves” - the setting intensifies his strained state.
- Q18** | **Correct Answer: B** Hydrofracture could cause abrupt events not captured by gradual models → projections may underestimate the speed and severity.

- Q19** | **Correct Answer: C** Business owners foster investment in the neighborhood's "well-being" - the logical noun for what communities care about.
- Q20** | **Correct Answer: B** "...temperatures above freezing." - a period ends the sentence; a new sentence follows with competing hypotheses.
- Q21** | **Correct Answer: D** Bloom would say Nussbaum is "overly optimistic" because empathy's biases may limit literature's ability to produce broad moral growth.
- Q22** | **Correct Answer: B** Goal: explain significance of the CO2 detection. udes the detection, its first-ever status, and Kiang's assessment.
- Q23** | **Correct Answer: D** IDPs could be used to "preserve" temperature-sensitive drugs - maintaining them without refrigeration.
- Q24** | **Correct Answer: A** "...the softer layers beneath, creating..." - comma introduces a participial phrase describing the result.
- Q25** | **Correct Answer: C** Bebop's complexity "resisted" casual imitation - Parker and Gillespie made jazz hard to copy.
- Q26** | **Correct Answer: D** "In other words" signals a restatement: the structure was a spatial encoding of social and cosmological relationships.
- Q27** | **Correct Answer: B** Goal: present Aldiss's argument and note it is contested. B does both - states his claim and cites the counterargument.
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Section 2: Math

Module 1 (22 Questions) - All Students

Time: 35 minutes

Q1 | Correct Answer: B

Let n be the number of novels and p the number of poetry collections. The equation is $14n + 9p = 100$. Testing the positive integer possibilities gives $n = 2$ and $p = 8$, since $14(2) + 9(8) = 100$. Therefore, the correct answer is B.

Q2 | Correct Answer: C Skill: Data Analysis (Ratios from Table) | Difficulty: Easy

Species per hectare: Marshfield: $34/12 \sim 2.83$, Cedar Lake: $45/18 = 2.50$, Pine Bog: $24/8 = 3.00$, Heron Flats: $58/25 = 2.32$. Pine Bog has the greatest ratio.

Q3 | Correct Answer: D Skill: Function Evaluation | Difficulty: Easy

$$f(5) = 3(5) + 7 = 15 + 7 = 22.$$

Q4 | Correct Answer: B Skill: Rate of Change | Difficulty: Easy

$$\text{Rate} = (39 - 15)/(6 - 0) = 24/6 = 4 \text{ gallons per minute.}$$

Q5 | Correct Answer: D Skill: Linear Equations | Difficulty: Easy

$$3(2x - 4) = 18 \rightarrow 6x - 12 = 18 \rightarrow 6x = 30 \rightarrow x = 5.$$

Q6 | Correct Answer: C Skill: Percentages | Difficulty: Easy

$$45\% \text{ of } 200 = 0.45 \times 200 = 90.$$

Q7 | Correct Answer: D Skill: Line of Best Fit | Difficulty: Easy

$$y = 8(6) + 40 = 48 + 40 = 88.$$

Q8 | Correct Answer: C Skill: Exponent Rules | Difficulty: Easy

$$(4x^3)(5x^2) = 20x^{(3+2)} = 20x^5.$$

Q9 | Correct Answer: A Skill: Geometry (Perimeter) | Difficulty: Easy

Let width = w . Length = $3w$. Perimeter = $2(w + 3w) = 2(4w) = 8w = 64$. $w = 8$.

Q10 | Correct Answer: C Skill: Ratios | Difficulty: Easy

Ratio 3:5, total parts = 8. Red = $(3/8)(40) = 15$.

Q11 | Correct Answer: D Skill: Slope | Difficulty: Easy

Slope = $(13 - 5)/(6 - 2) = 8/4 = 2$.

Q12 | Correct Answer: C Skill: Percentages (Reverse) | Difficulty: Medium

If price is reduced by 20%, sale price = 80% of original. $0.80 \times P = 56$. $P = 56/0.80 = 70$.

Q13 | Correct Answer: D Skill: Systems of Equations | Difficulty: Easy

Add equations: $2x + y + x - y = 11 + 1 \rightarrow 3x = 12 \rightarrow x = 4$. Then $y = 11 - 2(4) = 3$.

Q14 | Correct Answer: B Skill: Triangle Area | Difficulty: Easy

Area = $(1/2)(10)(7) = 35$.

Q15 | Correct Answer: D Skill: Linear Models | Difficulty: Easy

Distance = rate $\times$ time $\rightarrow d = 55t$.

Q16 | Correct Answer: B Skill: Mean | Difficulty: Easy

Sum = $5 \times 12 = 60$. Known sum = $8 + 10 + 14 + 16 = 48$. Fifth number = $60 - 48 = 12$.

Q17 | Correct Answer: C Skill: Cylinder Volume | Difficulty: Medium

$V = \pi(4^2)(20) = \pi(16)(20) = 320\pi$.

Q18 | Correct Answer: B Skill: Sets/Inclusion-Exclusion | Difficulty: Medium

At least one = French + Spanish - Both = $18 + 12 - 5 = 25$.

Q19 | Correct Answer: D Skill: Data Analysis (Table Comparison) | Difficulty: Medium

Differences: Parrotfish: $320 - 280 = 40$, Wrasse: $150 - 145 = 5$, Damselfish: $45 - 40 = 5$, Surgeonfish: $200 - 190 = 10$. Wrasse and Damselfish both have a difference of 5, making their weights most similar across habitats.

Q20 | Correct Answer: C Skill: Unit Conversion | Difficulty: Medium

$120 \text{ feet} \times 12 \text{ inches/foot} = 1,440 \text{ inches}$. $1,440/8 = 180 \text{ seeds}$.

Q21 | Correct Answer: 62 Skill: Algebraic Manipulation (Grid-In) | Difficulty: Easy

$4x + 7 = 31 \rightarrow 4x = 24$. $8x + 14 = 2(4x + 7) = 2(31) = 62$.

Q22 | Correct Answer: 2 Skill: Systems/Word Problem (Grid-In) | Difficulty: Medium

Let s = boxes of 6, t = boxes of 10. $s + t = 3$ and $6s + 10t = 22$. From first: $t = 3 - s$. Substitute: $6s + 10(3 - s) = 22 \rightarrow 6s + 30 - 10s = 22 \rightarrow -4s = -8 \rightarrow s = 2$.

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Section 2: Math

Module 2 - Easier Path (22 Questions)

Time: 35 minutes

- Q1 | Correct Answer: D - $3(4) - 5 = 12 - 5 = 7$.
- Q2 | Correct Answer: C - $0.60 \times 250 = 150$.
- Q3 | Correct Answer: D - $(7 + 12 + 5 + 8)/4 = 32/4 = 8$.
- Q4 | Correct Answer: B - $2x = 12$, $x = 6$.
- Q5 | Correct Answer: D - $3/24 = x/40 \rightarrow x = 120/24 = 5$.
- Q6 | Correct Answer: C - $12 \times 5 = 60$.
- Q7 | Correct Answer: A - 3 red out of 10 total = $3/10$.
- Q8 | Correct Answer: B - $3x + 12 - 2x = x + 12$.

- Q9** | **Correct Answer: C** - $80 \times 2.5 = 200$.
- Q10** | **Correct Answer: D** - $7 \times 50 = 350$.
- Q11** | **Correct Answer: B** - Slope = $12/4 = 3$.
- Q12** | **Correct Answer: A** - 25% of \$40 = \$10 discount. Sale price = \$40 - \$10 = \$30.
- Q13** | **Correct Answer: C** - Side = $36/4 = 9$. Area = $9^2 = 81$.
- Q14** | **Correct Answer: D** - $2x - 3 > 7 \rightarrow 2x > 10 \rightarrow x > 5$. The smallest listed integer greater than 5 is 6.
- Q15** | **Correct Answer: B** - The middle value of 7 ordered numbers is the 4th: 8.
- Q16** | **Correct Answer: A** - \$3 for first hour + \$2 for each remaining (h - 1) hours = $3 + 2(h - 1)$.
- Q17** | **Correct Answer: D** - Same y-coordinate, so distance = $|5 - 1| = 4$.
- Q18** | **Correct Answer: C** - $y = 35/5 = 7$. $y + 2 = 9$.
- Q19** | **Correct Answer: B** - $C = \pi \cdot d = 10\pi$.
- Q20** | **Correct Answer: 40** - $1.40 \times x = 56 \rightarrow x = 56/1.40 = 40$.
- Q21** | **Correct Answer: 1** - $3(4) - 2b = 10 \rightarrow 12 - 2b = 10 \rightarrow 2b = 2 \rightarrow b = 1$.
- Q22** | **Correct Answer: D** - $P(\text{blue}) = 3/8$. With replacement: $P(\text{both blue}) = (3/8)(3/8) = 9/64$.

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Section 2: Math

Module 2 - Harder Path (22 Questions)

Time: 35 minutes

Q1 | **Correct Answer: D Skill: No-Solution Systems**

For no solution, the lines must be parallel: same slope, different intercepts.

Rewrite $kx + 6y = 15$ as $y = -(k/6)x + 15/6$ and $4x + 3y = 9$ as $y = -(4/3)x + 3$.

Set slopes equal: $-k/6 = -4/3$, so $k = 8$. The intercepts, 2.5 and 3, differ. Therefore, there is no solution.

Q2 | **Correct Answer: A** The vertex form $f(x) = 2(x - 3)^2 + 5$ shows the minimum value is 5 (when $x = 3$, the squared term is 0).

Q3 | **Correct Answer: D** Standard form: $(x - h)^2 + (y - k)^2 = r^2$. Center $(-2, 4)$, radius 7: $(x + 2)^2 + (y - 4)^2 = 49$.

Q4 | **Correct Answer: C** Vertex x-coordinate $= -b/(2a) = -(-12)/(2 \cdot 3) = 12/6 = 2$.

Q5 | **Correct Answer: A** Doubling every 3 hours: $P(t) = 500(2)^{t/3}$.

Q6 | **Correct Answer: D** $(x^2 - 9)/(x - 3) = (x + 3)(x - 3)/(x - 3) = x + 3$.

Q7 | **Correct Answer: B** $\sin(A) = 5/13$, so opposite $= 5$, hypotenuse $= 13$. Adjacent $= \sqrt{169 - 25} = \sqrt{144} = 12$. $\cos(A) = 12/13$.

Q8 | **Correct Answer: A**
The profit function $P(x) = -2x^2 + 120x - 400$ opens downward, so its maximum occurs at the vertex. $x = -b/(2a) = 30$, and $P(30) = 1,400$. Therefore, the correct answer is A.

Q9 | **Correct Answer: D** $\log_2(x) = 5$, so $x = 2^5 = 32$.

Q10 | **Correct Answer: C** $h(t) = 0$: $-5t^2 + 30t = 0 \rightarrow -5t(t - 6) = 0 \rightarrow t = 0$ or $t = 6$. Returns at $t = 6$.

Q11 | **Correct Answer: A**
Exactly one real solution requires a discriminant of 0: $b^2 - 4(2)(8) = 0$, so $b^2 = 64$ and $b = \pm 8$. Of the answer choices, only -8 appears. Therefore, the correct answer is A.

Q12 | **Correct Answer: D** $g(x)$ undefined when denominator $= 0$: $x - 4 = 0 \rightarrow x = 4$.

Q13 | **Correct Answer: A** $R = 15\sqrt{16} = 15(4) = 60$ thousand dollars.

Q14 | **Correct Answer: C** $a^2 + 8^2 = 17^2 \rightarrow a^2 = 289 - 64 = 225 \rightarrow a = 15$.

Q15 | **Correct Answer: A** Among public transit users (65 total), 35 commute under 30 min. Probability = $35/65$.

Q16 | **Correct Answer: C** $h(1) = a(1) - 4(1) + 7 = a + 3 = 6 \rightarrow a = 3$.

Q17 | **Correct Answer: 7** $x^2 - 5x - 14 = 0 \rightarrow (x - 7)(x + 2) = 0 \rightarrow x = 7$ or $x = -2$. Positive solution: 7.

Q18 | **Correct Answer: D** 4 units left: x goes from 3 to $3 - 4 = -1$. 5 units up: y goes from -2 to $-2 + 5 = 3$. New vertex: $(-1, 3)$.

Q19 | **Correct Answer: B** Infinitely many solutions: second equation is a multiple of the first. $6x - 4y = k$. The first equation times 2: $6x - 4y = 14$. So $k = 14$.

Q20 | **Correct Answer: A** $48\pi = (1/3)\pi r^2(9) = 3\pi r^2$. Thus $r^2 = 16$ and $r = 4$.

Q21 | **Correct Answer: -15** $(2x + 3)(x - 5) = 2x^2 - 10x + 3x - 15 = 2x^2 - 7x - 15$. So $c = -15$.

Q22 | **Correct Answer: 36** $f(4) = 3(2^4) = 48$ and $f(2) = 3(2^2) = 12$.
fore, $f(4) - f(2) = 48 - 12 = 36$.
